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RESEARCH ARTICLE

Long-term ear-EEG monitoring of sleep – A case study during shift work

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Summary

The interest in sleep as a potential clinical biomarker is growing, but the standard method of sleep assessment, polysomnography, is expensive, time consuming, and requires a lot of expert assistance for both set-up and interpretation. To make sleep analysis more available both in research and in the clinic, there is a need for a reliable wearable device for sleep staging. In this case study, we test ear-electroencephalography. A wearable, where electrodes are placed in the outer ear, as a platform for longitudinal at-home recording of sleep. We explore the usability of the ear-electroencephalography in a shift work case with alternating sleep conditions. We find the ear-electroencephalography platform to be reliable both in terms of showing substantial agreement to polysomnography after long-time use (with an overall agreement, using Cohen's kappa, of 0.72) and by being unobtrusive enough to wear during night shift conditions. We find that fractions of non-rapid eye movement sleep and transition probability between sleep stages show great potential as sleep metrics when exploring quantitative differences in sleep architecture between shifting sleep conditions. This study shows that the ear-electroencephalography platform holds great potential as a reliable wearable for quantifying sleep "in the wild", pushing this technology further towards clinical adaptation.

KEYWORDS
ear-electroencephalography, electroencephalography, mHealth, sleep, wearables

1 | INTRODUCTION

It is of high clinical importance to measure sleep not only for diagnostic purposes but also as a progression marker in various deseases (Buxton & Marcelli, 2010). Reliable information about sleep architecture and its changes over time is critical if we want to fully use patient sleep as a clinical tool (Medic et al., 2017). Beyond patient care, long-term information on sleep can help expand our knowledge of the impact on sleep of everyday work- and living conditions including nighttime- or shift work.

When we quantify sleep patterns, polysomnography (PSG) is the gold-standard. The electroencephalography (EEG) electrodes of the

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PSG are essential to provide information on the sleep architecture. However, standard PSG setups are time consuming, expensive, and restricted to in-laboratory settings because the amount and placement of equipment makes it obtrusive and might in itself cause sleep disturbances (Newell et al., 2012). If we want to understand the impact of different sleep conditions on a larger population scale, devices that are accessible, cost-efficient and unobtrusive are required.

In recent years there has been an increasing focus on wearable platforms for such accessible, cost-efficient and unobtrusive sleep staging (Imtiaz, 2021). Wearable sleep devices are often based on actigraphy, which measures rest/activity cycles, often paired with one or more additional biomarkers such as pulse oximetry, temperature measurements, heart rate, heart variability and skin conductance (Imtiaz, 2021).

These devices are very user friendly as they can often be worn as a wristband with a low level of obtrusiveness. The element of unobtrusiveness is particularly relevant when devices are used for data collection over long periods of time (ultra-long-term monitoring).

Although actigraphy-based devices can often be used for ultra-long-term monitoring, the information they hold is limited. These non-EEG devices do not provide information about sleep architecture and, although some non-EEG wearables combine several other bio-signals (e.g. actigraphy, heart rate variability and skin conductance), the output can only represent proxy measurements of sleep stages.

The use and integration of non-EEG bio-signals for sleep staging might improve over time, but as of now sleep staging as four- or five-stage classification is poor based on these devices (de Zambotti et al., 2020).

There are multiple wearable EEG-based devices on the market, but they tend to focus on seizure detection in epilepsy (Nielsen et al., 2021), whereas sleep tracking is a less common focus. Some EEG wearable sleep platforms are on the market, often as headbands (de Zambotti et al., 2020).

These headbands are frequently marketed as biohacking wearables with a focus on feedback instead of sleep staging, and sleep is therefore often not recorded as the classical five-stage architecture (wake, rapid eye movement [REM], N1, N2 and N3) according to the AASM guidelines (Berry et al., 2015). A few headband devices do offer five-stage classification (Arnal et al., 2020; Finan et al., 2016 and Mota-Rolim et al., 2019); however, the designs of these headbands are often obtrusive, which limits the use of them to nighttime sleep settings.

An alternative to the headband-based devices is an ear-based EEG platform in which EEG is recorded from electrodes placed in or around the ear (Mikkelsen et al., 2015). One example is Sterr et al. (2018), who conducted a sleep study based on the so-called cEEGrid, which is an around-the-ear flex-printed electrode array. Other ear-EEG-based sleep studies have used in-the-ear electrodes, which has the advantage that it can easily be applied by the user without assistance.

The ear-EEG has shown promising results compared with PSG (Mikkelsen et al., 2017; Tabar et al., 2020; Zibrandtsen et al., 2016), and has previously been used as a 24/7 wearable device for up to 5 days in patients with epilepsy in a hospital setting (Zibrandtsen et al., 2017).

In this case study, we test ear-EEG as a platform for longitudinal at-home recording of sleep, and explore the quantitative differences in sleep architecture between normal nights and shift work nights.

We measure sleep architecture and examine the effect of shift work. A few studies using a full standard EEG or PSG in shift workers have been performed (Akerstedt, 2003; Akerstedt et al., 1991). These studies are based on more obtrusive techniques of full EEG or PSG, which may hinder the workflow and have therefore only been used for a few days. Separate studies have used a system of a single or two EEG electrodes (Mentink et al., 2020), but also limited to a few nights of observation.

In this study, we test the ear-EEG platform in a shift work setting, to see if it can be used as a possible future platform for ultra-long-term studying of sleep conditions.

2 | MATERIALS AND METHODS

2.1 | Methods

The study was conducted during the spring of 2020. It was an observational study with an explorative aim and therefore there were no predefined hypotheses.

In this study, we wanted to test the ear-EEG platform for longitudinal out-of-the-clinic recordings. We used a shift work protocol to inject a variance to the dataset, and looked at simple sleep metrices in normal night conditions and nights with abnormal conditions (shift work).

2.2 | Subject

The subject was a 29-year-old female who consented to participation in the study. The subject had no known medical problems and took no medication.

The subject slept alone in a darkened room on all nights. On work nights the subject slept at the hospital and had a work phone by the bed. On non-work nights the subject slept at home, and there was no phone or any other technical devices in the room. The subject did not use an alarm clock on any of the nights during the recording period. The subject knew the ear-EEG device and had worn it twice before during daytime testing periods, but had no experience with using the device during work or sleep periods.

2.3 | Ear-EEG

The ear-EEG devices used in this study consisted of two earpieces (one in each ear) with five dry-contact iridium-oxide electrodes (Kappel et al., 2018) embedded in each. The earpieces were placed in

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the outer portion of the external ear canal and the concha. Each earpiece was individualized by creating an impression mould using silicone (Otoform, Xpand, Detax, software, 2.0), which was used as a model for a 3D printed silicone earpiece. An air duct through the earpiece allows normal hearing.

The ear-EEG electrodes were connected to a wireless Mobita amplifier (TMS International, Ej Oldenzaal, the Netherlands) via a multi connector (ConfiCap, TMS International, Ej Oldenzaal, the Netherlands). Recordings were stored in the recording device with no inspection of the signal prior to or during recordings. The sampling rate was set to 1000 Hz.

No maintenance was performed on the ear-EEG device during the entire period of the trial. No electrodes were changed or repaired, and no parts of the equipment were cleaned. This was done to best simulate "in-the-wild" use of the device.

2.4 | Polysomnography (PSG)

The PSG setup was done using 10 EEG scalp electrodes (F3, F4, C3, C4, O1, O2, M1, M2, Fpz), two electrooculogram electrodes, three chin electrodes, four leg electrodes, two electrocardiogram (ECG) electrodes and two respiratory belts (thorax and abdomen).

Conductive gel was used for all PSG electrodes. Signals were visually inspected before each recording, and the acceptable electrode impedance upper limit was set at 5 kOhm.

For scoring, F4-M1, C4-M1 and O2-M1 were used as EEG channels, with F3, C3, O1 and M2 as backup. Fpz was used as ground.

For the PSG recording, a Trackit EEG amplifier was used (Trackit MK3, Lifelines Neuro Company, LLC, Louisville, KY, 40206 USA). For the nights with concurrent ear-EEG and PSG measurements, to prevent interference or influence between the measuring devices, there was no galvanic connection between the devices. Recordings were afterwards synchronized from a planned series of chewing and blinking artefacts.

2.5 | Recordings

Recordings using the ear-EEG device started in the evenings before the subject went to bed.

The subject had six nightshifts (five were successfully recorded) in a Covid-19 unit. On non-night shift days, the subject worked approximately 8 hr a day during the daytime.

During night shifts, the subject reported to the ward at 15:30 hours, and left at 09:00 hours the following morning. The subject was free to sleep during downtime. On these nights, the recordings started at some point prior to the time where the subject felt the urge to go to bed. The time when the subject went to bed (lights out) and the time when the subject woke up were logged in a sleep diary, as were all times out of bed after lights out. Phone calls that did not prompt the subject to leave the bed or turn the lights on were not logged.

On all the night shifts, the subject wore the ear-EEG device while doing a combination of deskwork, discussions with colleagues, patient examinations and phone calls.

Because the place of work was a Covid-19 unit, all patient exams were done using protective goggles, face masks and isolation gowns. All protective gear was single use, and therefore put on right before each patient exam.

Data were recorded over two periods. The first recording period was between 22 March and 30 April 2020. This 40-night period resulted in 30 nights of successful ear-EEG recordings. Of the missing 10 recordings, two were due to ear problems not related to the device, four were missed because the subject either forgot to record or forgot to recharge the device, three recordings were overwritten on the amplifiers memory card, and 1 night was not recorded due to discomfort probably caused by the ear-EEG device.

The second period was between 11 May and 15 May, and did not involve shift work. The intension was to record 4 nights with concurrent PSG and ear-EEG. Unfortunately, due to technical problems the PSG recordings of the first nights failed. A new PSG recording was set up at day 3 (14 May), resulting in 2 nights with concurrent PSG and ear-EEG. The ear-EEG device was used during all 5 nights.

This resulted in 39 nights where the test person wore the device. Of these, 36 nights were recorded but 1 night was cut short due to discomfort, leaving 35 nights of successfully recorded ear-EEG and 2 nights of corresponding PSG.

Figure 1 contains an overview of the 35 nights.

2.6 | Manual scoring

The data collected from the PSG were scored manually by a certified sleep technician who had vast experience with PSG-scoring (> 10,000 PSGs). All recordings were scored as 30-s epochs as either wake, N1, N2, N3 or REM according to the AASM guidelines. The sleep technician had no access to ear-EEG recordings or to the results from the scoring algorithm.

2.7 | Scoring algorithm

The scoring algorithm was used only on the ear-EEG data.

Each epoch was described by 83 statistical descriptors, so-called features, which were passed to a "random forest" classification algorithm, featuring 100 decision trees. For each classification task, features for seven consecutive epochs were passed to the algorithm (central epoch + three epochs to either side).

The algorithm was trained using a previously recorded data set of 80 recordings from 20 subjects (see Mikkelsen, Ebajemito, et al., 2019; Mikkelsen, Tabar, et al., 2019 for a detailed description

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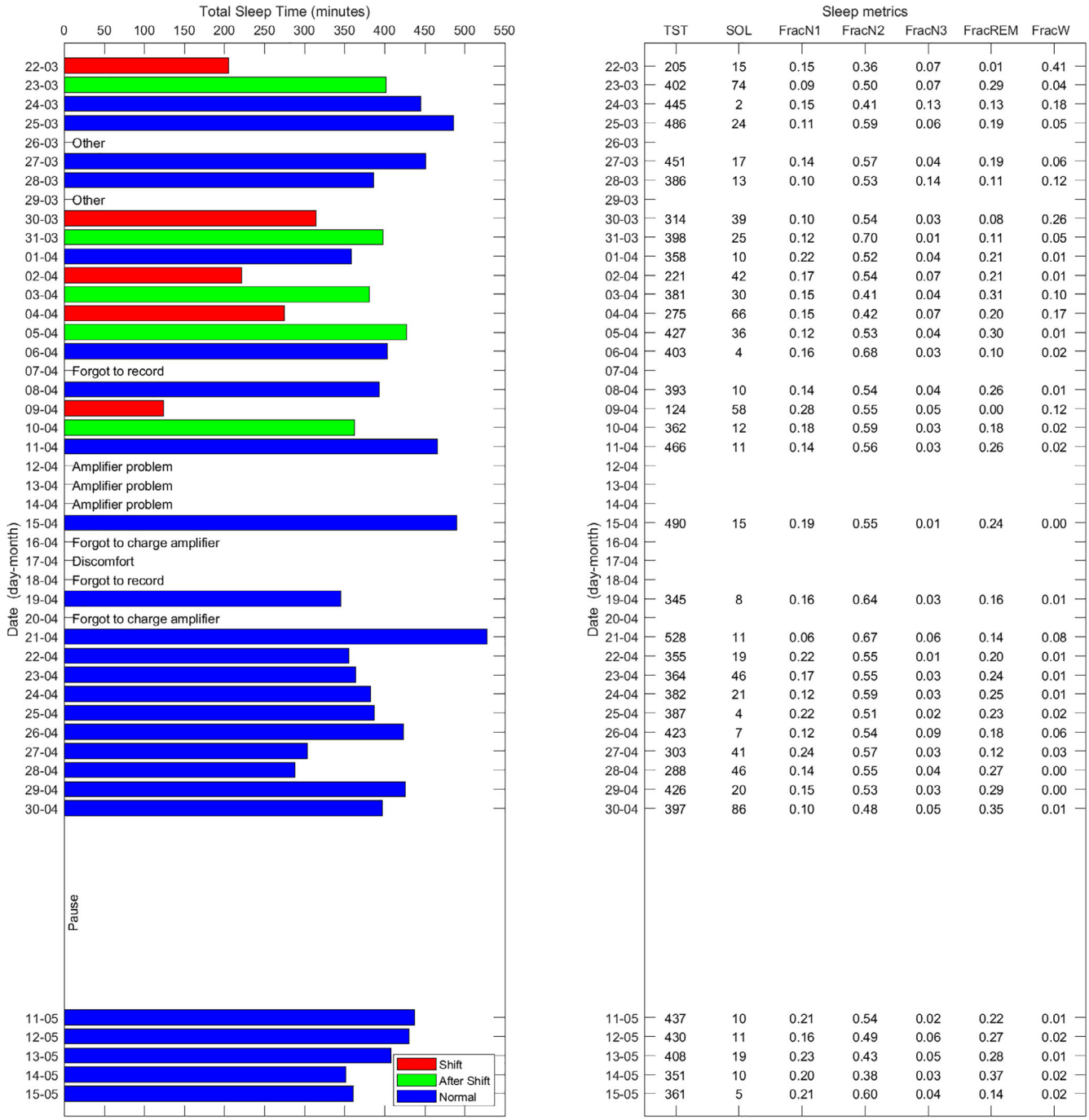

FIGURE 1 The left part of the figure illustrates the subjects use of the ear-electroencephalography (EEG) during the recording period. Red are nights where the subject was on shift, green marks nights following a shift night, and blue are normal nights. The *x*-axis depicts TST. The right part of the figure shows seven sleep metrices on corresponding nights. FracN1, the fraction of N1 sleep; FracN2, the fraction of N2 sleep; FracN3, the fraction of N3 sleep; FracREM, the fraction of rapid eye movement (REM) sleep; FracW, the fraction of wake during the sleep period; SOL, sleep-onset latency (min); TST, total sleep time (min)

of the training data, the features extracted and the general preprocessing pipeline).

Cohen's kappa was calculated for agreement between PSG and ear-EEG based on the recording period from "sleep onset" to "sleep end".

2.8 | Recording quality

As previously described, the ear-EEG had six electrodes in each ear. The signal was measured as the difference between the average of good electrodes in the left ear and the average of good electrodes

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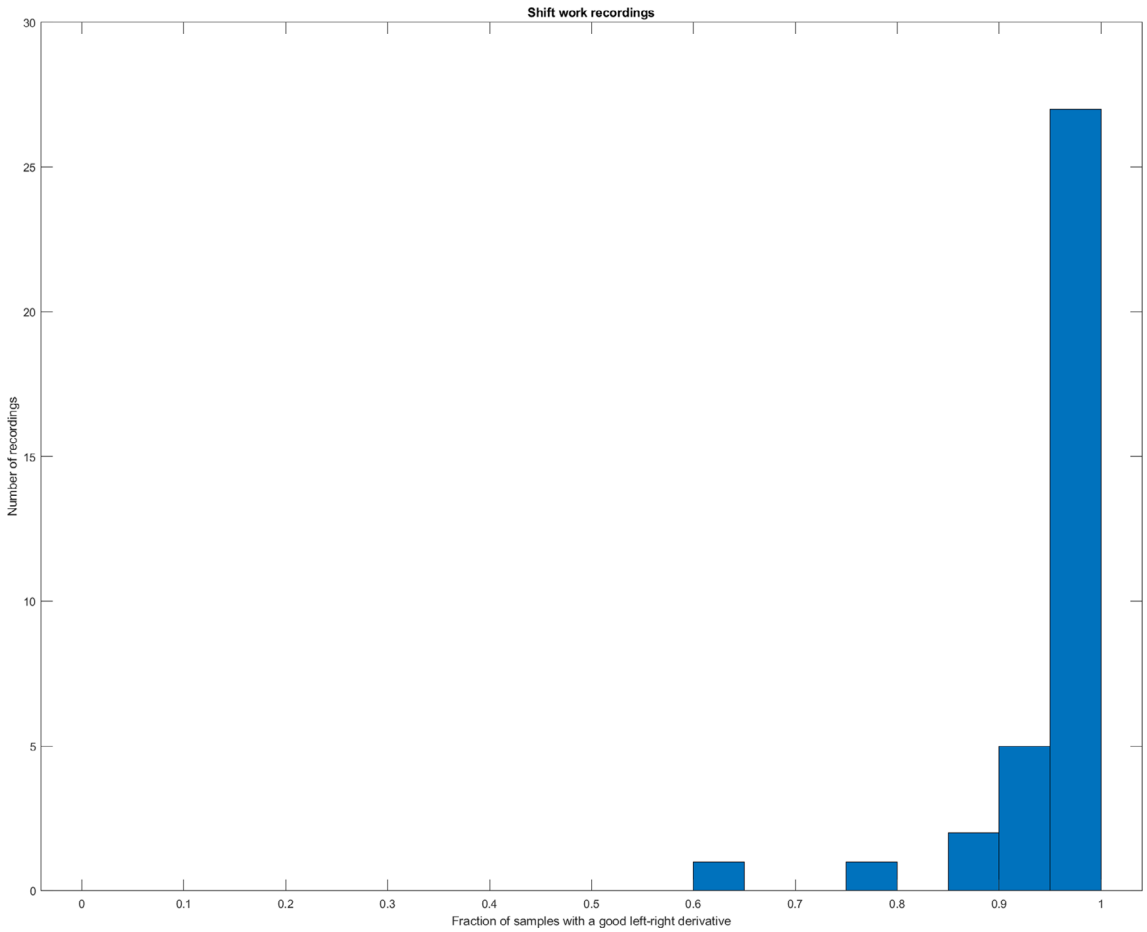

FIGURE 2 Histogram showing the distribution of the 36 recordings based on the data quality (rate of samples with good left–right derivative). The *x*-axis shows the fraction of non-rejected samples, and the *y*-axis shows the number of recordings

in the right ear; thus, a useful signal required at least one good electrode in each ear. Samples were rejected according to the rejection criteria outlined in Mikkelsen, Ebajemito, et al. (2019) and Mikkelsen, Tabar, et al. (2019), and all non-rejected samples were considered as good samples. We calculated the rate of good samples in each recording.

2.9 | Sleep metrics and regression model

To investigate if there were quantitative differences in the sleep architecture between normal-nights and shift-nights, basic sleep features were used as input for a logistic regression model (MATLAB 2021a, mnrfit, Statistics and Machine Learning Toolbox Version 12.1). To calculate the probability of a given night to be a normal night, the logistic regression model was trained on $N - 1$ nights, leaving out the night for which the probability was calculated. Thereby the model was trained *N*-times to get the probability for each of the *N*-recordings. A night was classified as a normal night if the output of the regression model was larger than 0.5.

Sleep onset was set at the first sleep epoch that was followed by at least nine consecutive sleep epochs (first 5 min of continuous sleep), and the end of the sleep period was set at the last epoch of the last 10 consecutive sleep epochs in the recording. This was done because the starting time of the recording relative to the time where the subject went to bed varied between normal nights and shift nights. A potential misclassification of a recorded wake-epoch before sleep onset would lead to a misinterpretation of both the total sleep time and the sleep architecture. Because recordings during shift nights were often started hours before lights out, such misclassification could potentially severely bias the outcome when comparing shift night with normal nights. We used sleep stage fractions and REM- and non-REM (NREM) transition probabilities for the logistic model, with N1 and N2 being pooled due to unreliable classification of N1. This resulted in three sleep fractions (fracN1N2, FracN3 and fracREM) and four transition probabilities between REM and NREM sleep stages ($p_{NREM\text{-}NREM}$, $p_{NREM\text{-}REM}$, $p_{REM\text{-}NREM}$, $p_{REM\text{-}REM}$). Notice that these seven features are all relative numbers between 0 and 1, and are not directly dependent on the total sleep time.

Finally, to describe differences between the normal-nights and the shift-nights we used the confusion matrix, and calculated the accuracy and the specificity of the regression model.

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3 | RESULTS

3.1 | Usability

Figure 1 shows the use of the ear-EEG device during the recording periods.

The ear-EEG device was usable without any type of maintenance from the beginning of the first recording period on 20 March to the end of the second period on 15 May. During this time, the subject wore the device for a total of 39 nights, but due to missing recordings only 36 recordings were filed.

3.2 | Data quality

Figure 2 shows the distribution of good left-to-right derivative samples for the 36 recordings from the ear-EEG.

In 75% of all recordings (27 of 36), > 95% of the data are of good quality. In 89% of all recordings (32 of 36; 89%), > 90% of the data are good.

3.3 | Algorithm agreement

To assess the quality of the ear-EEG-based automatic sleep scoring, the subject had 2 nights of recording with both ear-EEG and PSG.

Figure 3 shows a confusion matrix of the pooled epochs from the two manually scored PSG-nights versus the algorithm scored ear-EEG data from those same nights.

The overall agreement, using Cohen's kappa, was 0.72, indicating a substantial agreement between the algorithm scored ear-EEG data and the manually scored PSG.

The algorithm performed well for REM epochs where sensitivity and specificity are both 0.95. It was more likely to fail in its predictions when classifying N1 and N2 sleep, often misinterpreting one for the other. Most commonly N1 being classified as N2.

About one in four of N3 epochs are wrongly classified as N2.

Figure 4 shows examples of two algorithm scored hypnograms (one normal night [A] and one night shift night [B]). All hypnograms can be found in the supplementary material in Appendix S1.

3.4 | Regression model classifier

To explore quantitative differences in sleep architecture between normal nights and shift work nights, we used stage transition probabilities and the fractions of NREM2, NREM3 and REM sleep.

Nights were separated into three categories: "Shift night"; "After shift night" (the first night after a shift night); and "Normal night" (all nights not being a night of shift work or the first following night).

Inspecting the distribution of the seven sleep metrics in Figure 5, there were no apparent differences between normal-nights and shift-nights; however, when plotting each combination of metrics in scatterplots (see supplementary material in Appendix S1) we find that there seems to be a tendency for the centre of mass for night shift nights to drift from the centre of mass for normal nights. To explore this further, we used the metrics for a logistic regression model and performed leave-one-out cross-validation.

Figure 6 shows the result of the classification model.

The classifier labels 24 of 25 normal nights and four of five shift nights correctly with classification based on sleep stage fractions (NREM2, NREM3 and REM) and transitions between REM and NREM.

The classifier performs with an accuracy of 93% and a specificity of 80%.

Manual Label \ Aut. Label	W	R	N1	N2	N3	?	Sensitivity / Specificity
W	72 5%	1 0%	15 1%	4 0%	0 0%	0 0%	0.90 0.99
R	1 0%	445 28%	5 0%	51 3%	0 0%	0 0%	0.95 0.95
N1	7 0%	6 0%	28 2%	101 6%	0 0%	0 0%	0.48 0.93
N2	0 0%	14 1%	9 1%	374 24%	0 0%	0 0%	0.58 0.98
N3	0 0%	0 0%	1 0%	111 7%	333 21%	0 0%	1.00 0.91
?	0 0%	0 0%	0 0%	0 0%	0 0%	0 0%	NaN 1.00

FIGURE 3 Confusion matrix with manually scored epochs from polysomnography (PSG) on the vertical axis, and algorithm predicted scoring of epochs from the ear-electroencephalography (EEG) on the horizontal axis. The sixth column and row represent epochs that could not be classified. The top number is the number of epochs, and the bottom number is the percentage of all epochs. Sensitivity and specificity for each stage are shown

4 | DISCUSSION

Ear-EEG has emerged as a platform for long-term monitoring of sleep, and has previously shown substantial agreement with PSG. In this study, ear-EEG was used for longitudinal "in-the-wild" sleep recordings.

4.1 | Relation to previous studies

The ear-EEG platform has previously been tested as a method for sleep monitoring. A case study on a single night from one healthy subject in 2016 (Zibrandtsen et al., 2016) demonstrated the potential of the platform. This was followed by the introduction of an automated

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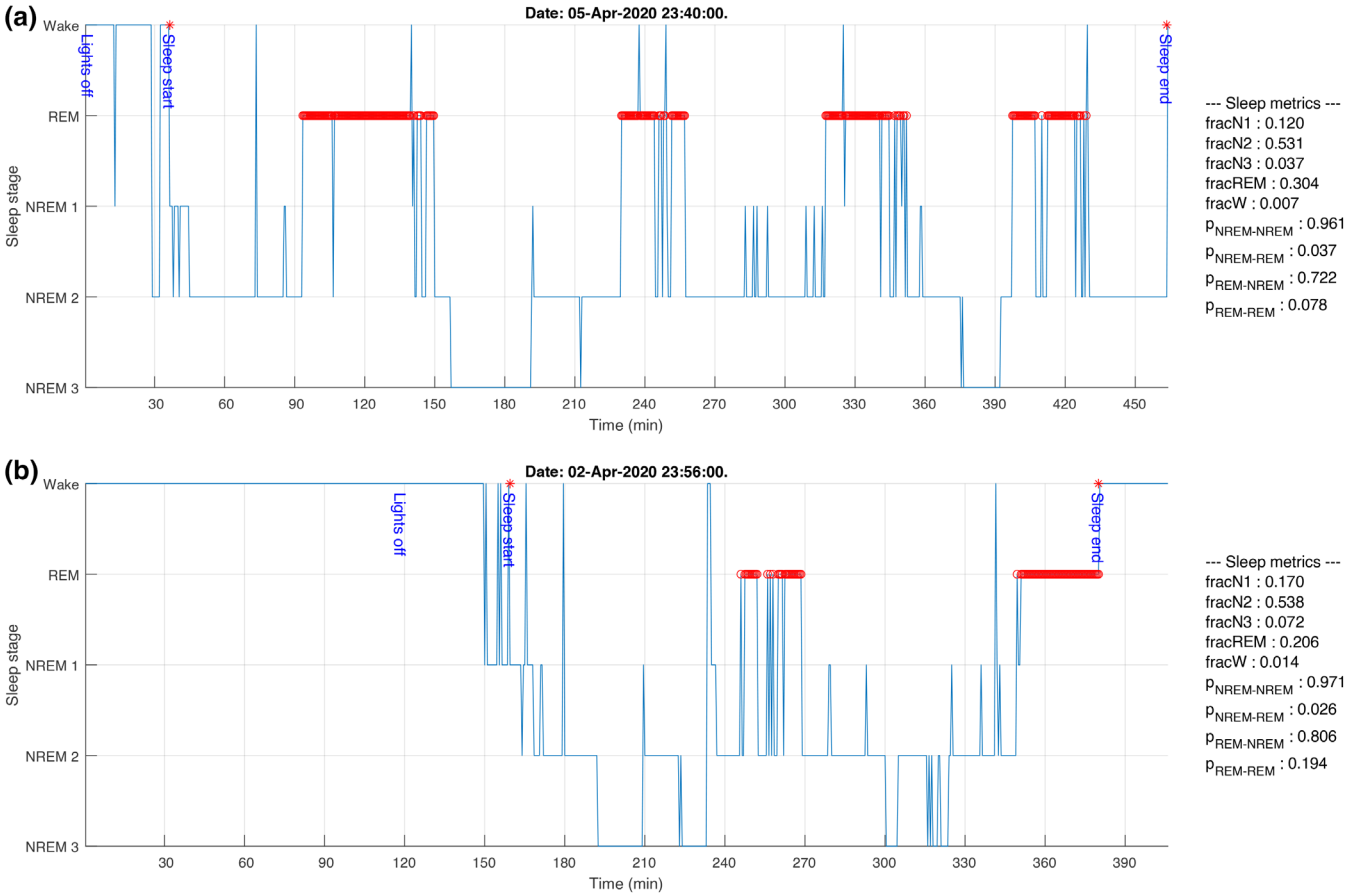

FIGURE 4 Example of hypnograms. (a) Hypnogram from a normal night. (b) Hypnogram from a night shift: *x*-axis, time in minutes; *y*-axis, sleep stages (non-rapid eye movement [NREM]1, non-REM stage N1; NREM2, non-REM stage 2; NREM3, non-REM stage). Sleep start marks the start of the sleep period used for the classifier (first epoch of the first 5 min of consistent sleep); Sleep end marks the end of the sleep period used for the classifier (last epoch of the last 5 min of consistent sleep). REM sleep epochs are marked in red. Note that the *x*-axis is not aligned between (a) and (b)

classifier (Stochholm et al., 2016) used to compare 7741 epochs of automatically classified ear-EEG with manually scored PSG finding an overall kappa value of up to 0.68 for five-stage classification (Mikkelsen et al., 2017).

In 2019, 27,593 epochs of manually scored ear-EEG were compared with scalp EEG, in a population of patients with epilepsy, in a hospital setting where patients wore the devise for up to 5 days. The overall kappa value was 0.74 (Joergensen et al., 2020).

In 2019, 20 healthy subjects slept for 4 nights with ear-EEG, an actigraph and PSG in a home setting (with daily checks of the equipment). The kappa value was up to 0.72 for five-stage classification, significantly outperforming actigraphy (Mikkelsen, Ebajemito, et al., 2019; Mikkelsen, Tabar, et al., 2019; Tabar et al., 2020). In 2022, the ear-EEG was tested in 10 subjects who self-applied the device, showing that recordings are still reliable when subjects apply the equipment without assistance (Mikkelsen et al., 2022).

In this study, we used the platform in an "in-the-wild" setting for the first time. We showed that the ear-EEG device is still capable of performing after being worn for 39 nights over a period of 47 days without any type of checks or maintenance, suggesting that the ear-EEG is a robust device and has potential as a home monitoring platform.

4.2 | Using the ear-EEG platform

When evaluating a sleep-monitoring device, unobtrusiveness and sleep scoring accuracy are critical features.

The subject decided to remove the device on 1 night due to discomfort. It might be that the in-ear design of the device takes some time to get used to or that wearing it for a long time can cause periods of discomfort. This of course poses a problem as some potential users might choose not to wear it again after experiencing discomfort. In a previous study, 13 patients wore the ear-EEG for 24 hr a day for up to 5 days. On 10% of the nights, a patient had removed the device due to discomfort (Joergensen et al., 2020). Since then, the earpieces have gone from rigid acrylic casts with wet electrodes to the softer silicone pieces with dry electrodes used in this study. Further design changes are likely needed to further improve the comfort.

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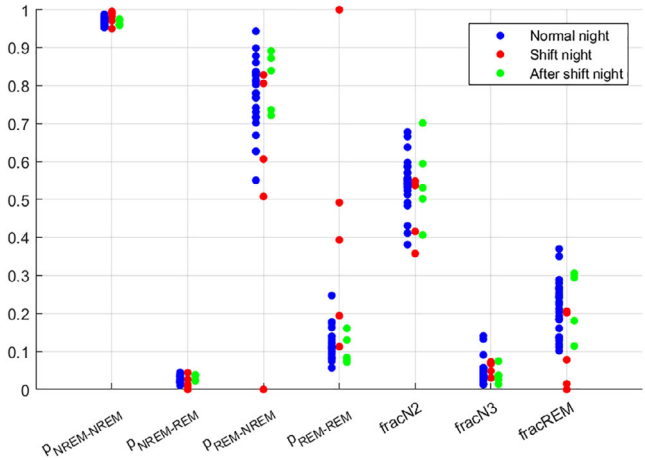

FIGURE 5 Plot of the seven sleep metrices used in the logistic regression model. $P_{\text{NREM-NREM}}$, probability of the next epoch being a non-rapid eye movement (NREM) epoch when coming from a NREM epoch; $P_{\text{NREM-REM}}$, probability of the next epoch being a REM epoch when coming from a NREM epoch; $P_{\text{REM-NREM}}$, probability of the next epoch being a NREM epoch when coming from a REM epoch; $P_{\text{REM-REM}}$, probability of the next epoch being a REM epoch when coming from a NREM epoch; fracN2, the fraction of N2 sleep from sleep start to sleep end; fracN3, the fraction of N3 sleep from sleep start to sleep end; fracREM, the fraction of REM sleep from sleep start to sleep end. Note that the first four metrics are probabilities and the last three are fractions, the *y*-axis is therefore not comparable between the two types of metrics

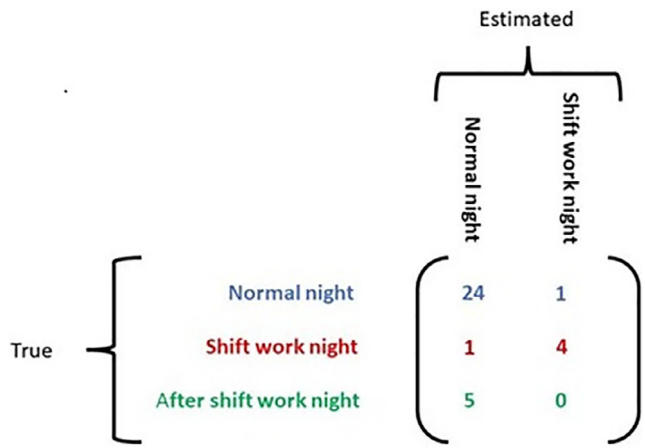

FIGURE 6 Confusion matrices for logistic regression and leave-one-out cross-validation, using both fraction of sleep stages and transitions between rapid eye movement (REM) and non-rapid eye movement (NREM) sleep

The subject in this study used the ear-EEG on five nightshifts in a busy Covid-19 unit, during a variety of tasks including patient examinations using protective gear and rest/wake transitions. The subject found no need to disconnect or remove the device at any time, indicating a good level of unobtrusiveness.

4.3 | Comparison to other EEG-based wearables

Most wearable sleep trackers are non-EEG devices using actigraphy often coupled with one or more metrics that reflect the autonomic nervous system (e.g. heart rate variability), but there are EEG-based wearables for sleep tracking presented in the literature, often as headbands (de Zambotti et al., 2020). These consumer devices rarely offer five-stage classification, but often group sleep stages into wake, REM, light sleep and deep sleep, which makes it difficult to compare these devices with the ear-EEG.

Of those offering five-stage classification, we identified the Aurora headband (iWinks LLC; Mota-Rolim et al., 2019), the Dreem headband (Dreem; Arnal et al., 2020) and Sleep Profiler (Advanced Brain Monitoring; Finan et al., 2016). We were not able to find peer-reviewed studies on the Aurora headband.

The Dreem headband is recording EEG, accelerometry and pulse oximetry, and using an automated classifier to label sleep stages. It has been validated against PSG with 25 healthy participants sleeping 1 night in a sleep laboratory. Validation showed a kappa value of 0.75 when compared with PSG, with agreement rates above 80% for REM, N2 and N3 but below 50% for N1 (Arnal et al., 2020).

The Sleep Profiler is recording EEG and accelerometry. In a study from 2016, 14 healthy participants slept for 1 night in a sleep laboratory with the Sleep Profiler and a PSG setup. Agreements were above 80% for REM, above 70% for N2 and N3, but below 30% for N1 (Finan et al., 2016).

The cEEGrid, behind-the-ear platform, was tested with five-stage classification in a paper published in 2019 (Mikkelsen, Ebajemito, et al., 2019; Mikkelsen, Tabar, et al., 2019) and performed with a kappa value of up to 0.58.

We find a kappa value of 0.72 when comparing the automatically classified ear-EEG data with manually scored PSG. This is comparable to previous ear-EEG studies (Joergensen et al., 2020; Mikkelsen et al., 2017; Tabar et al., 2020; Zibrandtsen et al., 2016). Like the two headbands, we find that N1 is the most difficult stage for the automated classifier to label correctly as it is often mistaken for N2. The distinction between N1 and N2 is also the most difficult when testing interrater variability between manual scores (Younes et al., 2016), which also highlights the problem with using potentially mislabelled epochs when training automatic scoring algorithms based on manual scoring.

4.4 | Classification regression model

The subject in this study used the ear-EEG device while on a shift work schedule that included 5 nights of nighttime work (18-hr shift). Previous studies have found that restricted sleep time results in a decrease in total sleep time (TST), an increase in the relative amount of slow-wave sleep (SWS) and a decrease in the relative amount of REM sleep (Akerstedt, 2003; Akerstedt et al., 1991; Ong et al., 2016; Mentink et al., 2020).

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In leave-one-out cross-validation, 24 out of 25 normal nights and four out of five shift nights were classified correctly. The first night following a shift night was always classified as a normal night.

The limited number of nights in this study resulted in using the simplest form of the transition matrix (REM versus NREM), but with a larger dataset this could be expanded to separate NREM sleep stages, which may lead to an improvement in classification accuracy.

4.5 | Using the ear-EEG platform in night shift conditions

Shift work is very common, yet we have limited knowledge of its effect on the brain physiology and sleep. According to one study (Wickwire et al., 2017), shift-workers make up approximately 20% of the workforce in industrialized nations in health professions.

Several studies link night shift work with wide-ranging health issues like diabetes (Knutsson & Kempe, 2014), cancer (Kecklund & Axelsson, 2016) and cardiovascular diseases (Vyas et al., 2012). The effect of night shift work on brain function is most often evaluated via subjective reports in questionnaires of stress, sleep and well-being, via psychophysics measures of reaction time or cognitive function or via peripherally located devices like actigraphy or ECG. When EEG has been measured it has mostly been done in a laboratory setting due to the difficulties of wearing traditional EEG equipment during work (Kang et al., 2020; Mansukhani et al., 2012; Vanttola et al., 2018).

Ambulatory PSG has only been reported in few cases and only with a few nights of recording in each subject. Being able to collect ultra-long-term data from shift workers could help our understanding of the impact that shift work has on sleep patterns.

This study serves as a pilot project for the use of the ear-EEG platform to reliably obtain and classify sleep data when used for long-term recordings in an out-of-the-lab setting. It shows that ear-EEG can be used to record sleep-EEG in shift work conditions.

In future studies on the impact of shift work on sleep rhythms and sleep architecture, a mobile EEG device like the ear-EEG seems like a logical next step, as long-term PSG while working is impractical, and sleep diaries, questionnaires and actigraphy cannot provide the same information. In a larger population of shift workers, a wearable like the ear-EEG with automated classification could be used to verify potential problematic sleep architecture.

4.6 | Limitations

This is a case study that examines the use of the ear-EEG platform by examining nights from the same subject. We cannot conclude that the results we find here can be transferred to a larger population.

Our subject is a young adult and, although we consider the ear-EEG platform easy to use, age might have an impact on compliance. Future studies will have to expand the test population both in terms of size and demographics.

It should be noticed that the ear-EEG used in this study is a prototype and further research in the practical usability of the wearable might result in further improvements of the device.

For the comparison between the automatically classified ear-EEG data and the manually scored PSG we only had 2 nights of sleep, which may not be enough to be representative. We only had one sleep technician scoring the two PSGs and therefore we cannot consider interrater variability into our results.

4.7 | Conclusion

In this study, the ear-EEG platform is used "in-the-wild" for the first time and in a setting of shifting sleep conditions. We use the platform as a means of long-term EEG monitoring of 47 nights without any checks or maintenance. We find the ear-EEG platform to be reliable both in terms of showing substantial agreement to PSG after long-time use and by being unobtrusive enough to wear during night shift conditions.

AUTHOR CONTRIBUTIONS

SDJ: collected all data, wrote most of the manuscript, and did part of the data analyses and figures. PK: did most of the data analyses and part of the figures. MB: helped write the manuscript. KM: did part of the data analysis and figures. MCH: did most of the pre-processing of the data. MLR: did part of the pre-processing of the data. TWK: initiated the project and helped write the manuscript. All authors edited and approved the final manuscript.

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CONFLICT OF INTEREST STATEMENT

Authors MCH and MLR are employed by T&W Engineering A/S who developed the ear-EEG device. TWK is used as a consultant for T&W. All other authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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